LEVEL 1 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

PRACTICE ASSESSMENT 2 (FSM109P)

MARK SCHEME

Section A	Process (Task description)	Total mark	Mark allocation	Comments	PS or US	Subject content
Question 1	Correct calculation using positive and negative numbers	**1**	**1 mark:** Correct answer, ie -7		US	2
Question 2	Correct multiplication by 1000	**1**	**1 mark**: Correct answer, ie 150		US	3a
Question 3	Correct subtraction of decimals	**1**	**1 mark:** Correct subtraction, ie (45.53 – 24.37) = 21.16		US	11b
Question 4	Correct answer using order of operations	**1**	**1 mark**: Correct answer, ie 55		US	7
Question 5	Correct equivalent percentage shown	**1**	**1 mark**: Correct percentage, ie 40%		US	16a
Question 6	Correct order of fractions	**1**	**1 mark:** Correct order, ie $\frac{1}{2}$ $\frac{3}{5}$ $\frac{2}{3}$ $1\frac{1}{5}$ $1\frac{1}{4}$	Accept largest to smallest	US	8

Question 7	Correct price of one item Correct price of all 3 items Method to find total price Correct total price	**4**	**1 mark**: Correct price of one item, eg (39 ÷ 100) = 0.39 OR 39(p) OR (18.50 ÷ 10) = 1.85 OR (14.29 ÷ 10) = 1.429 OR 1.43	Accept 1.42	PS	3b
			1 mark: Correct price for all three items, ie (£)0.39 OR 39p AND (£)1.85 AND (£)1.429 OR 1.43 OR 1.42		PS	3b
			1 mark: Method to find total price of items, eg 2 × 0.39 + 1.85 + 3 × 1.43 = (£6.92)	Accept use of 1.42 OR 1.429	PS	3b
			1 mark: Correct price, ie (£)6.91 OR (£)6.92	Accept (£)6.89 Do not accept more than 2dp	PS	3b
Question 8	Method for finding other side of ratio Correct number of ml and decision	**2**	**1 mark**: Method for finding correct amount of paint, eg 180 ÷ 3 × 2 (= 120) OR 100 ÷ 2 × 3 (= 150)		PS	17a
			1 mark: Correct number of ml, ie 120 (ml) OR 150 (ml) AND 'No'		PS	17a
Question 9	Evidence of using estimation Correct method to find perimeter of 1 painting Valid estimation of amount of tape needed for 5 paintings	**3**	**1 mark**: Valid values for estimation, eg 50 (cm) OR 30 (cm) OR 49cm OR 33 (cm)	Accept any valid estimation values	PS	15
			1 mark: Valid method for finding perimeter, eg 50 + 50 + 30 + 30 (= 160cm) OR 49 + 49 + 33 + 33 (= 164cm) OR	Accept correct perimeter from non-estimated values, ie 163cm	PS	22b
			1 mark: Correct estimation of amount of tape needed for 5 paintings, eg (5 × 160) = 800 (cm)	Do not accept non-estimated answer	PS	15

<table>
<tr><th>Section B</th><th>Process (Task description)</th><th>Total mark</th><th>Mark allocation</th><th>Comments</th><th>PS or US</th><th>Subject content (SoS)</th></tr>
<tr><td>Question 10</td><td>Identify plan of 3d shape</td><td>1</td><td>1 mark: Correct answer, ie
D (Cuboid)</td><td></td><td>US</td><td>25a</td></tr>
<tr><td>Question 11</td><td>Complete grouped frequency table</td><td>1</td><td>1 mark: Correctly completed frequency column, ie
<table><tr><th>Age (years)</th><th>Frequency</th></tr><tr><td>0 – 9</td><td>5</td></tr><tr><td>10 – 19</td><td>3</td></tr><tr><td>20 – 29</td><td>4</td></tr><tr><td>30 – 39</td><td>4</td></tr><tr><td>40 – 49</td><td>2</td></tr></table></td><td>Allow tally or totals.</td><td>US</td><td>28a</td></tr>
<tr><td>Question 12</td><td>Select highest percentage</td><td>1</td><td>1 mark: Correct % chosen, ie
87%</td><td></td><td>US</td><td>13</td></tr>
<tr><td>Question 13</td><td>Method to find volume

Correct volume found

Method to find number of balls

Correct number of balls found</td><td>4</td><td>1 mark: Method to find volume, ie
2.4 × 2.4 × 0.45 = (2.592m³)</td><td></td><td>PS</td><td>23</td></tr>
<tr><td></td><td></td><td></td><td>1 mark: Correct volume, ie
2.592 (m³)</td><td></td><td>PS</td><td>23</td></tr>
<tr><td></td><td></td><td></td><td>1 mark: Method to find number of balls, ie
2.592 × 2250</td><td>FT for incorrect volume</td><td>PS</td><td>23</td></tr>
<tr><td></td><td></td><td></td><td>1 mark: Correct number of balls found, ie
5832</td><td></td><td>PS</td><td>23</td></tr>
<tr><td>Question 14</td><td>Method to find number of black mats

Correct number of black, light grey and dark grey mats

Grid filled in with at least one line</td><td>3</td><td>1 mark: Correct method to find number of black mats, ie
30 ÷ 5 = (6)</td><td></td><td>PS</td><td>9</td></tr>
<tr><td></td><td></td><td></td><td>1 mark: Correct number of black, light grey and dark grey mats, ie
6 black, 12 light grey, 12 dark grey</td><td>May be seen on diagram</td><td>PS</td><td>9</td></tr>
<tr><td></td><td></td><td></td><td>1 mark: Pattern completed with at least one line of symmetry</td><td>Allow FT for incorrect numbers of mats as long as at least two colours used.</td><td>PS</td><td>24a</td></tr>
</table>

	of symmetry					
Question 15	Method to find cost of food or party bags Method to find total cost Correct total cost before discount Method to find 15% Correct final cost after discount	**5**	**1 mark**: Method to find cost of food OR party bags, eg 30 ÷ 5 × 17 (= £102) OR 6 × 17 (= £102) OR 30 ÷ 10 x 12 (=£36) OR 3 x 12 (=£36)		PS	17b
			1 mark: Method to find total cost of party, eg 102 + (3 × 12) + 90 = (228)	Allow FT for incorrect cost of food or party bags	PS	17b
			1 mark: Correct answer, ie (£)228		PS	17b
			1 mark: Valid method to find 15% or 85%, eg 15/100 × 228 = (34.2) OR 0.85 × 228 = (193.8) OR Any other valid method	FT from their total cost	PS	19
			1 mark: Correct answer, ie (228 – 34.2) = (£)193.80		PS	19
Question 16	Correct number written in words	**1**	**1 mark**: 832304 written correct in words, ie Eight hundred and thirty two thousand, three hundred and four	Ignore spelling mistakes and use of 'and'.	US	1
Question 17	Correct probability	**1**	**1 mark**: Correct probability, ie 3/10		US	31
Question 18	Calculate range	**1**	**1 mark**: Correct range, ie (25.8 – 19.1) = 6.7		US	29b
Question 19	Correct substitution into formula Correct upper and lower heart rates	**3**	**1 mark**: Correct substitution into formula for either upper or lower heart rate, ie 220 – 36 × 0.6 = (110.4) OR 220 – 36 × 0.7 = (128.8)		PS	5
			1 mark: Correct upper and lower heart rates, ie 110.4 AND 128.8		PS	5

	Correct decision		**1 mark**: Correct decision, ie Yes	Must be supported by calculations	PS	5
Question 20	Method to calculate total time	**2**	**1 mark**: – Method to add time, eg 45 + 45 + 75 + 20 × 5 = (265 (mins)) OR ¾ + ¾ + 1 ¼ + 1:40 = (2:45 + 1:40) OR any other valid method	Award if 265 or 4h 25m seen	PS	20e
	Correct total time in hours and minutes		**1 mark**: Correct total time in hours and minutes, ie 4 hours 25 minutes		PS	20e
Question 21	Method to find number of metres	**4**	**1 mark**: Method to find total number of metres 6680 × 0.7 = (4676m)	Award if 4676 and 4.676 seen	PS	20a
	Correct number of metres		**1 mark**: Correct number of metres, ie 4676 (m)	Award if 4.676 seen	PS	20a
	Conversion of m to km		**1 mark:** Conversion of metres to kilometres, ie 4.676 (km)	FT from their number of metres	PS	20a
	Rounded answer to 1 dp		**1 mark:** Correct number of km rounded to 1 dp, ie 4.7 (km)	FT from their number of km	PS	12b
Question 22	Correct numbers chosen	**3**	**1 mark:** Evidence that correct numbers chosen to find mean, eg 113 +121 + 112 + 118 + 115 = (579)		PS	29a
	Method to find mean		**1 mark:** Method to find mean, ie 579 ÷ 5 = (115.8)		PS	29a
	Correct mean found		**1 mark:** Correct mean, ie (579 ÷ 5 =) 115.8		PS	29a

	Process (Task description)	Total mark	Mark allocation	Comments	PS or US	Subject content
Question 23	Appropriate scale given. Data points at correct heights Graph appropriately labelled	**3**	**1 mark**: Appropriate scale given		US	27b
			1 mark: Data points at correct height (tolerance plus/minus 1 division) and line drawn between	Do not award for bar chart	US	27b
			1 mark: Graph contains appropriate axis labels and title, eg × axis: Weeks Y axis: Number of parcels Title: Graph to show number of parcels delivered over 4 weeks.	Accept similar wording for axis labels and title.	US	27b
Question 24	Method to calculate number of hours Correct number of hours Method to calculate number of hours holiday Correct number of hours holiday	**4**	**1 mark**: Valid method to calculate the number of hours, eg 2304.26 ÷ 8.72 (= 264.25)		PS	11d
			1 mark: Correct number of hours, ie 264.25		PS	11d
			1 mark: Correct method to calculate number of hours holiday, ie 264.25 × 0.12 = (31.71)	Accept 264 × 0.12 Allow FT for their number of hours	PS	11c
			1 mark: Correct number of hours, ie 31.71 (hours) OR 31 hours 41 or 42 minutes OR 31 (hours) OR 32 (hours)	Accept 31.68	PS	11c
Question 25	Correct amount put in savings Method to find 5% Correct amount of	**3**	**1 mark**: Correct amount to be saved, ie (2304.26 – 399) ÷ 2) = 952.63		PS	18
			1 mark: Method to calculate 5%, eg 0.05 × 952.63= (47.6315) OR 5 ÷ 100 × 952.63 = (47.6315) OR 1.05 × 952.63 = (£1000.26) OR Any other valid method	FT through from their amount put in savings.	PS	18
			1 mark: Correct amount in savings account after 1		PS	18

	savings after interest added		year, ie (952.63 + 47.63) = (£)1000.26			
Question 26	Correct value for one missing length	**5**	**1 mark**: Correct value for missing length, eg (57-25) = 32 (m) OR (62 – 35) = 27 (m)		PS	22a
	Method to calculate area		**1 mark**: Valid method to calculate area of composite shape, eg 25 × 62 + 32 × 35 = (2670) OR 62 × 57 - 27 × 32 = (2670) OR Any other valid method	Award if 2670 seen FT for incorrect missing lengths	PS	22a
	Correct area		**1 mark:** Correct area, ie 2670 (m²)		PS	22a
	Method to calculate number of trees		**1 mark:** 2670 ÷ 64 (= 41.718…)	FT for incorrect missing lengths	PS	22a
	Correct number of trees		**1 mark**: Correct number of trees, ie 41	Do not accept decimal answer	PS	22a

Annotation notes:

Annotation	Meaning
US	Underpinning skills
PS	Problem solving skills
FT	Follow through
(…)	Information that is not required for the mark point

Functional Skills in Mathematics Level 1 – Mapping matrix

Paper number	**RFSMO109**					
Task number	**Section A**		**Section B**		**Total**	**%**
Total number of marks per Section	**15**		**45**			
Problem Solving (PS) maximum marks **Underpinning skills (US) maximum marks**	**9** **6**		**36** **9**		**Total no of sub-elements mapped = 31**	
Tick the box to confirm that Section B contains at least three 4 – 7 mark questions.			✓✓✓			
Level 1 Subject Content	**PS**	**US**	**PS**	**US**		
1. Read and write order and compare large numbers (up to one million)				1(Q16)	1	
2. Use both positive and negative numbers		1(Q1)			1	
3a. Multiply whole numbers and decimals by 10, 100, 1000		1(Q2)			1	
3b. Divide whole numbers and decimals by 10, 100, 1000	4(Q7)				4	
4. Use multiplication facts and make connections with division facts						
5. Use simple formulae expressed in words for one or two-step operations			3(Q19)		3	
6. Calculate the squares of one-digit and two-digit numbers						
7. Follow the order of precedence of operators		1(Q4)			1	
8. Read, write, order and compare common fractions and mixed numbers		1(Q6)			1	
9. Find fractions of whole number quantities or measurements			2(Q14)		2	
10. Read, write order and compare decimals up to three decimal places						
11a. Add decimals with decimals up to two decimal places						
11b. Subtract decimals with decimals up to two decimal places		1(Q3)			1	
11c. Multiply decimals with decimals up to two decimal places			2(Q24)		2	
11d. Divide decimals with decimals up to two decimal places			2(Q24)		2	
12a. Approximate by rounding to a whole number			1(Q21)		1	

12b. Approximate by rounding to one or two decimal places				1(Q12)	1	
13. Read, write order and compare percentages in whole numbers						
14. Calculate percentages of quantities, including simple percentage increases and decreases by 5% and multiples thereof						
15. Estimate answers to calculations using fractions and decimals	2(Q9)				2	
16a. Recognise equivalences between common fractions, percentages and decimals		1(Q5)			1	
16b. Calculate equivalences between common fractions, percentages and decimals						
17a. Work with simple ratio	2(Q8)				2	
17b. Work with direct proportions			3(Q15)		3	
Total: Number and number system	**PS**	**US**	**PS**	**US**	**29**	
18. Calculate simple interest in multiples of 5% on amounts of money			3(Q25)		3	
19. Calculate discounts in multiples of 5% on amounts of money			2(Q15)		2	
20a. Convert between units of length in the same system			3(Q21)		3	
20b. Convert between units of weight in the same system						
20c. Convert between units of capacity in the same system						
20d. Convert between units of money in the same system						
20e. Convert between units of time in the same system			2(Q20)		2	
21. Recognise and make use of simple scales on maps and drawings						
22a. Calculate the area of simple shapes including those that are made up of a combination of rectangles			5(Q26)		5	
22b. Calculate the perimeter of simple shapes including those that are made up of a combination of rectangles	1(Q9)				1	
23. Calculate the volumes of cubes and cuboids			4(13)		4	
24a. Draw 2-D shapes and demonstrate an understanding of line symmetry			1(14)		1	
24b. Understand the relative size of angles						
25a. Interpret plans and elevations of simple 3-D shapes				1(Q10)	1	

25b. Interpret nets of simple 3-D shapes						
26. Use angles when describing position and direction and measure angles in degrees						
Total: Measure, shape and space	**PS**	**US**	**PS**	**US**	**22**	
27a. Represent discrete data in tables and diagrams						
27b. Represent discrete data in charts i) pie charts, ii) bar charts and iii) line graphs				3(Q23)	3	
28a. Group discrete data				1(Q11)	1	
28b. Represent grouped data graphically						
29a. Find the mean of a set of quantities			3(Q22)		3	
29b. Find the range of a set of quantities				1(Q18)	1	
30. Understand probability on a scale from 0 (impossible) to 1 (certain) and use probabilities to compare the likelihood of events						
31. Use equally likely outcomes to find the probabilities of simple events and express them as fractions				1(Q17)	1	
Total: Handling data	**PS**	**US**	**PS**	**US**	**9**	
Total Mark PS/US Total %	9	6	36	9		